

Technical report
Sparsifying l_2 regularized least squares regression
by biconjugate convex relaxation

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March 13, 2026

Contents

0.1	Linear least squares	1
0.2	Sparsifying linear least squares	1
0.3	Relaxation $(\mathbb{Q}^{\beta,\delta})$	2
0.3.1	Optimality conditions for $(\mathbb{Q}^{\beta,\delta})$	2
0.3.2	Dual of $(\mathbb{Q}^{\beta,\delta})$	3
0.3.3	Theoretical links with Ridge and Lasso	5
0.3.4	Mixing roles: $(\mathbb{Q}^{\beta,\delta})$ vs $(\text{ELASTICNET}^{\beta',\gamma})$	7
0.4	Solving $(\mathbb{Q}^{\beta,\delta})$	8
0.4.1	Cyclic coordinate descent for solving $(\mathbb{Q}^{\beta,\delta})$	12
0.4.2	Interior point method	13
0.4.3	Branch and bound method	13
0.5	Sparse Principal Component	14

0.1 Linear least squares

$$(\text{RIDGE}^\beta) \begin{cases} \min & \|Ax - b\|_2^2 + \beta \|x\|_2^2 \\ \text{s.t.} & x \in \mathbb{R}^n, \end{cases} \quad (1)$$

where $\beta > 0$ is a parameter.

0.2 Sparsifying linear least squares

We use a l_0 regularization approach to promote sparse solutions as expressed in the following problem

$$\begin{cases} \min & \|Ax - b\|_2^2 + \rho \|x\|_0 + \beta \|x\|_2^2 \\ \text{s.t.} & x \in \mathbb{R}^n. \end{cases} \quad (2)$$

where $\rho \geq 0$ is a parameter which controls the sparsity of the solutions.

With $\delta := \sqrt{\frac{\rho}{\beta}}$, we have $\rho = \beta\delta^2$. The previous problem becomes

$$(\mathbb{P}^{\beta,\delta}) \begin{cases} \min & \|Ax - b\|_2^2 + \beta\delta^2 \|x\|_0 + \beta \|x\|_2^2 \\ \text{s.t.} & x \in \mathbb{R}^n. \end{cases} \quad (3)$$

0.3 Relaxation $(\mathbb{Q}^{\beta,\delta})$

Replacing $\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2$ by its tightest convex under approximation, its biconjugate,

$$\left(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2 \right)^{**} (x),$$

see Le Thi et al., we obtain the following biconjugate relaxation (**BiConjugate Relaxation, BCR**)

$$(\mathbb{Q}^{\beta,\delta}) \begin{cases} \min & F(x) := \|Ax - b\|_2^2 + \beta\delta^2 \left(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2 \right)^{**} (x) \\ \text{s.t.} & x \in \mathbb{R}^n. \end{cases}$$

$(\mathbb{Q}^{\beta,\delta})$ is a convex under approximation of $(\mathbb{P}^{\beta,\delta})$ and $(\mathbb{Q}^{\beta,\delta})$ solves $(\mathbb{P}^{\beta,\delta})$ when its solution satisfies particular global optimality conditions (see Le Thi et al.).

$$\left(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2 \right)^{**} (x) = \sum_i^n \left(|\cdot|_0 + \frac{1}{\delta^2} |\cdot|_2^2 \right)^{**} (x_i),$$

$$\left(|\cdot|_0 + \frac{1}{\delta^2} |\cdot|_2^2 \right)^{**} (x_i) = \frac{1}{\delta^2} x_i^2 - \frac{1}{\delta^2} \left[(\delta - |x_i|)^+ \right]^2 + 1 = \begin{cases} \frac{1}{\delta^2} x_i^2 + 1 & \text{if } |x_i| \geq \delta. \\ 2\frac{1}{\delta} |x_i| & \text{otherwise.} \end{cases}$$

0.3.1 Optimality conditions for $(\mathbb{Q}^{\beta,\delta})$

The following form for the optimality conditions are very useful for the theoretical study of the links between the biconjugate relaxation BCR, Ridge, Lasso and ElasticNet.

Optimality condition for $(\mathbb{Q}^{\beta,\delta})$

The optimality condition for the convex relaxation $(\mathbb{Q}^{\beta,\delta})$ is given by, Le Thi et al.

$$-2A^T(Ax - b) \in \beta\delta^2 \partial \left(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2 \right)^{**} (x).$$

As

$$\partial \left(|\cdot|_0 + \frac{1}{\delta^2} |\cdot|_2^2 \right)^{**} (x_i) = \begin{cases} \left\{ 2\frac{1}{\delta^2} x_i \right\} & \text{if } |x_i| \geq \delta, \\ \left\{ 2\frac{1}{\delta} \text{sign}(x_i) \right\} & \text{if } 0 < |x_i| \leq \delta, \\ \left[-2\frac{1}{\delta}, 2\frac{1}{\delta} \right] & \text{if } x_i = 0, \end{cases}$$

the condition becomes

$$\left(\mathbb{Q}_{opt}^{\beta,\delta}\right) [A^T(Ax - b)]_i \in \begin{cases} \{-\beta x_i\} & \text{if } |x_i| \geq \delta \\ \{-\delta\beta \operatorname{sign}(x_i)\} & \text{if } 0 < |x_i| \leq \delta \\ [-\delta\beta, \delta\beta] & \text{if } x_i = 0. \end{cases}$$

or

$$[A^T(Ax - b) + \beta x]_i \in \begin{cases} \{0\} & \text{if } |x_i| \geq \delta \\ \{\beta x_i - \delta\beta \operatorname{sign}(x_i)\} & \text{if } 0 < |x_i| \leq \delta \\ [-\delta\beta, \delta\beta] & \text{if } x_i = 0. \end{cases}$$

Proposition 0.3.1.1 $x = 0$ will be an optimal solution for $(\mathbb{Q}^{\beta,\delta})$ if and only if $\|A^T b\|_\infty \leq \delta\beta$.

Proof: Direct use of $(\mathbb{Q}_{opt}^{\beta,\delta})$. □

Proposition 0.3.1.2 If x is a solution for $(\mathbb{Q}^{\beta,\delta})$ and $\delta < \min_{i \in \operatorname{supp}(x)} |x_i|$, then x is a solution for $(\mathbb{Q}^{\beta,\delta'})$, for all $\delta', \delta \leq \delta' \leq \min_{i \in \operatorname{supp}(x)} |x_i|$.

Proof: Direct use of $(\mathbb{Q}_{opt}^{\beta,\delta})$. □

0.3.2 Dual of $(\mathbb{Q}^{\beta,\delta})$

Lagrangian dual of $(\mathbb{D}^{\beta,\delta})$ is

$$(\mathbb{D}^{\beta,\delta}) \begin{cases} \max & G(\nu) := -\frac{1}{4} \|\nu\|_2^2 - b^T \nu - \beta\delta^2 \left(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2 \right)^* \left(-\frac{A^T \nu}{\beta\delta^2} \right) \\ \text{s.t.} & \nu \in \mathbb{R}^m. \end{cases}$$

With

$$\left(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2 \right)^* (z) = \sum_i^m \left(|\cdot|_0 + \frac{1}{\delta^2} |\cdot|_2^2 \right)^* (z_i),$$

$$\left(|\cdot|_0 + \frac{1}{\delta^2} |\cdot|_2^2 \right)^* (z_i) = \left(\frac{\delta^2}{4} z_i^2 - 1 \right)^+ = \begin{cases} \frac{\delta^2}{4} z_i^2 - 1 & \text{if } |z_i| \geq \frac{2}{\delta} \\ 0 & \text{otherwise.} \end{cases}$$

Weak duality holds. As Slater's condition is satisfied, strong duality holds for $(\mathbb{D}^{\beta,\delta})$.

Optimality condition for $(\mathbb{D}^{\beta,\delta})$

The optimality condition for the convex program $(\mathbb{D}^{\beta,\delta})$ is given by,

$$\frac{1}{2}\nu + b \in A\partial(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2)^*\left(-\frac{A^T\nu}{\beta\delta^2}\right).$$

As

$$\partial(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2)^*(z_i) = \begin{cases} \left\{\frac{\delta^2}{2}z_i\right\} & \text{if } |z_i| > \frac{2}{\delta} \\ \{0\} & \text{if } |z_i| < \frac{2}{\delta} \\ [0, \delta] & \text{if } z_i = \frac{2}{\delta}, \\ [-\delta, 0] & \text{if } z_i = -\frac{2}{\delta}, \end{cases}$$

the condition becomes

$$\left(\mathbb{D}_{opt}^{\beta,\delta}\right) \frac{1}{2}\nu + b = Ay, \quad y_i \in \begin{cases} \left\{-\frac{1}{2\beta}[A^T\nu]_i\right\} & \text{if } |[A^T\nu]_i| > 2\delta\beta \\ \{0\} & \text{if } |[A^T\nu]_i| < 2\delta\beta \\ [0, \delta] & \text{if } [A^T\nu]_i = -2\delta\beta, \\ [-\delta, 0] & \text{if } [A^T\nu]_i = 2\delta\beta. \end{cases}$$

Finding a dual feasible from a primal feasible

The previous primal and dual optimality conditions show that if x is a primal solution then $\nu = Ax - b$ is a dual solution.

Therefore for an arbitray feasible primal point x we define a dual feasible point $\nu = s\nu_0$, $\nu_0 = (Ax - b)$. The scalar s will be the argmax of the following 1D convex program:

$$\begin{cases} \max & -\frac{1}{4}\|\nu_0\|_2^2 s^2 - b^T\nu_0 s - \beta\delta^2 \left(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2\right)^*\left(-\frac{A^T\nu_0}{\beta\delta^2}s\right) \\ s.t. & s \in \mathbb{R}, \end{cases}$$

say

$$\begin{cases} \max & -\frac{1}{4}\|\nu_0\|_2^2 s^2 - b^T\nu_0 s - \beta\delta^2 \sum_{i=1}^n \left(\frac{[A^T\nu_0]_i^2}{4\delta^2\beta^2} s^2 - 1\right)^+ \\ s.t. & s \in \mathbb{R}. \end{cases}$$

This is a piecewise 1D quadratic convex program. A solution will have necessarily the opposite sign of $b^T\nu_0$.

We will first solve

$$\begin{cases} \max & -\frac{1}{4}\|\nu_0\|_2^2 s^2 + |b^T\nu_0|s - \beta\delta^2 \sum_{i=1}^n \left(\frac{[A^T\nu_0]_i^2}{4\delta^2\beta^2} s^2 - 1\right)^+ \\ s.t. & s \geq 0, \end{cases}$$

and then update the sign of s .

Duality gap

$G(\nu)$ lower bounds $F(x)$ and the corresponding duality gap

$$\eta := F(x) - G(\nu)$$

is always nonnegative from weak duality. x is no more than η -suboptimal. At the optimum the gap is zero from strong duality.

0.3.3 Theoretical links with Ridge and Lasso

Theoretical links with Ridge

Recall

$$(\text{RIDGE}^\beta) \begin{cases} \min & \|Ax - b\|_2^2 + \beta \|x\|_2^2 \\ \text{s.t.} & x \in \mathbb{R}^n. \end{cases} \quad (4)$$

Ridge optimality condition $(2A^T(Ax - b) + 2\beta x = 0)$ can be written as

$$\left(\text{RIDGE}_{opt}^\beta \right) [A^T(Ax - b)]_i \in \{-\beta x_i\}.$$

Proposition 0.3.3.1 *A RIDGE is always a $(\mathbb{Q}^{\beta, \delta})$.*

More precisely, assume A has no null column,

1. *If x is a solution for (RIDGE^β) ,
then x is a solution for $(\mathbb{Q}^{\beta, \delta})$ for all $\delta \leq \min_i |x_i|$.*
2. *If x is a solution for $(\mathbb{Q}^{\beta, \delta})$ and $\delta \leq \min_i |x_i|$,
then x is a solution for (RIDGE^β) .*

Proof: Direct use of $\left(\text{RIDGE}_{opt}^\beta \right)$ and $\left(\mathbb{Q}_{opt}^{\beta, \delta} \right)$. □

Theoretical links with Lasso

$$(\text{LASSO}^\gamma) \begin{cases} \min & \|Ax - b\|_2^2 + \gamma \|x\|_1 \\ \text{s.t.} & x \in \mathbb{R}^n. \end{cases} \quad (5)$$

Lasso optimality condition can be written as

$$\left(\text{LASSO}_{opt}^\gamma \right) [A^T(Ax - b)]_i \in \begin{cases} \left\{ -\frac{\gamma}{2} \text{sign}(x_i) \right\} & \text{if } x_i \neq 0 \\ \left[-\frac{\gamma}{2}, \frac{\gamma}{2} \right] & \text{if } x_i = 0. \end{cases}$$

Proposition 0.3.3.2 *A LASSO is always a $(\mathbb{Q}^{\beta, \delta})$.*

More precisely,

1. If x is a solution for $(LASSO^\gamma)$,
then x is a solution for $(\mathbb{Q}^{\frac{\gamma}{2\delta}, \delta})$ for all $\delta \geq \max_{i \in \text{supp}(x)} |x_i|$.
2. If x is a solution for $(\mathbb{Q}^{\beta, \delta})$ and $\delta \geq \max_{i \in \text{supp}(x)} |x_i|$,
then x is a solution for $(LASSO^{2\delta\beta})$.

Proof: Direct use of $\left(\text{LASSO}_{opt}^\gamma\right)$ and $\left(\mathbb{Q}_{opt}^{\beta, \delta}\right)$. □

From Ridge to Lasso. How navigate $(\mathbb{Q}^{\beta, \delta})$

Consider (RIDGE^β) , for a $\beta > 0$. (Assume A has no null column)
Define

$$\delta_{\text{RIDGE}^\beta} := \min_{i \in \text{supp}(x^{\text{RIDGE}^\beta})} |x_i^{\text{RIDGE}^\beta}|.$$

Let us analyze δ s values.

We already have x^{RIDGE^β} solution of $(\mathbb{Q}^{\beta, \delta})$ for all $\delta \leq \delta_{\text{RIDGE}^\beta}$.

We will consider $\delta_{\text{RIDGE}^\beta} > \delta$.

From inequality $F(x^{\mathbb{Q}^{\beta, \delta}}) \leq F(0)$, we deduce

$$|x_i^{\mathbb{Q}^{\beta, \delta}}| \leq \|x^{\mathbb{Q}^{\beta, \delta}}\|_1 \leq \frac{1}{2\delta\beta} F(0), \forall i.$$

So it suffices to constrain $\frac{1}{2\delta\beta} F(0) \leq \delta$ to ensure $|x_i^{\mathbb{Q}^{\beta, \delta}}| \leq \delta$, for all i .

So $x^{\mathbb{Q}^{\beta, \delta}}$ is a solution of $(LASSO^{2\delta\beta})$, for all $\delta > \max \left\{ \sqrt{\frac{1}{2\beta} F(0)}, \delta_{\text{RIDGE}^\beta} \right\}$.

Define

$$\delta_{\text{RIDGE}^\beta, \text{LASSO}} := \inf \left\{ \delta > \delta_{\text{RIDGE}^\beta} : x^{\mathbb{Q}^{\beta, \delta}} \text{ is a solution of } (LASSO^{2\delta\beta}) \right\}.$$

We also have necessarily

$$\rho_{\text{RIDGE}^\beta, \text{LASSO}} \leq \max \left\{ \sqrt{\frac{1}{2\beta} F(0)}, \delta_{\text{RIDGE}^\beta} \right\}.$$

Proposition 0.3.3.3 1. When δ moves from $\delta_{\text{RIDGE}^\beta}$ to $\delta_{\text{RIDGE}^\beta, \text{LASSO}}$,
 $(\mathbb{Q}^{\beta, \delta})$ moves from *RIDGE* to *LASSO*.

- before $\delta_{\text{RIDGE}^\beta}$, $(\mathbb{Q}^{\beta, \delta})$ is a *RIDGE*,
- after $\delta_{\text{RIDGE}^\beta, \text{LASSO}}$, $(\mathbb{Q}^{\beta, \delta})$ is a *LASSO*.

2. When δ moves strictly between $\delta_{\text{RIDGE}^\beta}$ and $\delta_{\text{RIDGE}^\beta, \text{LASSO}}$, $(\mathbb{Q}^{\beta, \delta})$ cannot be a RIDGE nor a LASSO.

The components of $x^{\mathbb{Q}^{\beta, \delta}}$ satisfy $\left(\mathbb{Q}^{\beta, \delta}_{\text{opt}}\right)$

$$\left[A^T(Ax^{\mathbb{Q}^{\beta, \delta}} - b)\right]_i \in \begin{cases} \left\{-\beta x_i^{\mathbb{Q}^{\beta, \delta}}\right\} & \text{if } \left|x_i^{\mathbb{Q}^{\beta, \delta}}\right| \geq \delta \\ \left\{-\delta \beta \text{sign}(x_i^{\mathbb{Q}^{\beta, \delta}})\right\} & \text{if } 0 < \left|x_i^{\mathbb{Q}^{\beta, \delta}}\right| \leq \delta \\ [-\delta \beta, \delta \beta] & \text{if } x_i^{\mathbb{Q}^{\beta, \delta}} = 0. \end{cases}$$

The second point of the previous proposition highlights the mixing role (on the components of x) played by $(\mathbb{Q}^{\beta, \delta})$. The first line of the optimality condition is for RIDGE, with β , and the followings are for LASSO, with $\gamma := 2\delta\beta$. The value $\delta = \frac{\gamma}{2\beta}$ defines a trade-off line on the coordinates value.

0.3.4 Mixing roles: $(\mathbb{Q}^{\beta, \delta})$ vs $(\text{ELASTICNET}^{\beta', \gamma})$

Theoretical links with ElasticNet

$$(\text{ELASTICNET}^{\beta', \gamma}) \begin{cases} \min & \|Ax - b\|_2^2 + \beta' \|x\|_2^2 + \gamma \|x\|_1 \\ \text{s.t.} & x \in \mathbb{R}^n. \end{cases} \quad (6)$$

ElasticNet optimality condition can be written as

$$\left(\text{ELASTICNET}^{\beta', \gamma}_{\text{opt}}\right) \left[2A^T(Ax - b)\right]_i \in \begin{cases} \{-2\beta' x_i - \gamma \text{sign}(x_i)\} & \text{if } x_i \neq 0 \\ [-\gamma, \gamma] & \text{if } x_i = 0. \end{cases}$$

Proposition 0.3.4.1 x is a common solution for $(\text{ELASTICNET}^{\beta', \gamma})$ and $(\mathbb{Q}^{\beta, \delta})$ if and only if x is a solution of $(\text{ELASTICNET}^{\beta', \gamma})$ or $(\mathbb{Q}^{\beta, \delta})$ and one of the following conditions holds:

1. there exists C_1 , such that

$$|x_i| = C_1, \forall i, x_i \neq 0,$$

$$C_1 \geq \delta, \beta = \beta' + \frac{\gamma}{2C_1}.$$

2. there exists C_2 , such that

$$|x_i| = C_2, \forall i, x_i \neq 0,$$

$$C_2 \leq \delta, \delta\beta = \frac{\gamma}{2} + \beta' C_2.$$

3. there exist C_1 and C_2 , $C_2 \leq C_1$, such that

$$|x_i| = C_1 \text{ or } C_2, \forall i, x_i \neq 0,$$

$$\delta = \frac{C_1(\gamma + 2\beta' C_2)}{\gamma + 2\beta' C_1}, \delta\beta = \frac{\gamma}{2} + \beta' C_2.$$

Proof: Use $\left(\text{ELASTICNET}_{opt}^{\beta', \gamma}\right), \left(\mathbb{Q}_{opt}^{\beta, \delta}\right)$ and compare solutions. $\square$

0.4 Solving $(\mathbb{Q}^{\beta, \delta})$

Let $\beta > 0$ and let us denote by $x^\beta(\delta)$ a solution for $(\mathbb{Q}^{\beta, \delta})$. The solution set with respect to δ is called the path and can be expressed as

$$\text{PATH}^\beta := \{x^\beta(\delta) : \delta \geq 0\}.$$

Computing a new solution on the path from a known one

Here we will consider that we have $(\mathbb{Q}^{\beta, \underline{\delta}})$, $\underline{\delta} > 0$ and a corresponding solution $\underline{x} := x^\beta(\underline{\delta})$.

Let us define the following partition of the indices i , $i = 1, \dots, n$.

$$\begin{aligned} \mathbb{K}_0 &:= \{i : |\underline{x}_i| > \underline{\delta}\}, \\ \mathbb{K}_1 \cup \mathbb{K}_2 &:= \{i : |\underline{x}_i| = \underline{\delta}\}, \mathbb{K}_1 \cap \mathbb{K}_2 = \emptyset, \\ \mathbb{K}_3 &:= \{i : 0 < |\underline{x}_i| < \underline{\delta}\}, \\ \mathbb{K}_4 \cup \mathbb{K}_5 &:= \{i : \underline{x}_i = 0, |[A^T(A\underline{x} - b)]_i| = \underline{\delta}\beta\}, \mathbb{K}_4 \cap \mathbb{K}_5 = \emptyset, \\ \mathbb{K}_6 &:= \{i : \underline{x}_i = 0, 0 < |[A^T(A\underline{x} - b)]_i| < \underline{\delta}\beta\}, \\ \mathbb{K}_7 &:= \{i : \underline{x}_i = 0, [A^T(A\underline{x} - b)]_i = 0\}, \\ \underline{M} &:= \text{diag}(\underline{\theta}), \\ \underline{\theta}_i := \text{sign}(\underline{x}_i) &:= \begin{cases} +1 & \text{if } \underline{x}_i > 0, \\ -1 & \text{if } \underline{x}_i < 0, \\ -\text{sign}([A^T(A\underline{x} - b)]_i) & \text{if } \underline{x}_i = 0, [A^T(A\underline{x} - b)]_i \neq 0, \\ +1 & \text{otherwise.} \end{cases} \end{aligned}$$

Let $\tau \in \{-1, +1\}$ given and consider the following polyhedra

$$\begin{array}{llllll}
(A^T \underline{AM})_i u + \beta \underline{\theta}_i u_i & & & & = (A^T b)_i & i \in \mathbb{K}_0 \\
(A^T \underline{AM})_i u + \beta \underline{\theta}_i u_i & & & & = (A^T b)_i & i \in \mathbb{K}_1 \\
(A^T \underline{AM})_i u + \beta \underline{\theta}_i u_i & & + \beta \underline{\theta}_i t_i & & = (A^T b)_i & i \in \mathbb{K}_2 \\
(A^T \underline{AM})_i u + \beta \underline{\theta}_i u_i & & + \beta \underline{\theta}_i t_i & & = (A^T b)_i & i \in \mathbb{K}_3 \\
(A^T \underline{AM})_i u + \beta \underline{\theta}_i u_i & & + \beta \underline{\theta}_i t_i & & = (A^T b)_i & i \in \mathbb{K}_4 \\
(A^T \underline{AM})_i u & + \beta \underline{\theta}_i s_i & & & = (A^T b)_i & i \in \mathbb{K}_5 \\
(A^T \underline{AM})_i u & + \beta \underline{\theta}_i s_i & & & = (A^T b)_i & i \in \mathbb{K}_6 \\
(A^T \underline{AM})_i u & - \beta \underline{\theta}_i s_i & + \beta \underline{\theta}_i w_i & & = (A^T b)_i & i \in \mathbb{K}_7 \\
u_i & + \tau \gamma & & - t_i & = \underline{\delta} & i \in \mathbb{K}_0 \\
u_i & + \tau \gamma & & - t_i & = \underline{\delta} & i \in \mathbb{K}_1 \\
u_i & + \tau \gamma & & + t_i & = \underline{\delta} & i \in \mathbb{K}_2 \\
u_i & + \tau \gamma & & + t_i & = \underline{\delta} & i \in \mathbb{K}_3 \\
u_i & + \tau \gamma & & + t_i & = \underline{\delta} & i \in \mathbb{K}_4 \\
& \tau \gamma & + s_i & & + w_i & = \underline{\delta} & i \in \mathbb{K}_5 \\
& \tau \gamma & + s_i & & + w_i & = \underline{\delta} & i \in \mathbb{K}_6 \\
& \tau \gamma & + s_i & & + w_i & = \underline{\delta} & i \in \mathbb{K}_7 \\
u_i & & & & = 0 & i \in \mathbb{K}_5 \\
u_i & & & & = 0 & i \in \mathbb{K}_6 \\
u_i & & & & = 0 & i \in \mathbb{K}_7
\end{array}$$

$$u \geq 0 \quad \gamma \geq 0 \quad s \geq 0 \quad t \geq 0 \quad w \geq 0 \quad .$$

The polyhedra contains $u := |\underline{x}|$, $\gamma := 0$, ... so is nonempty.

Proposition 0.4.0.1 $x := \underline{M}u$ satisfies $(\mathbb{Q}^{\beta, \underline{\delta} - \tau \gamma})$ optimality conditions.

Proof: With $x := \underline{M}u$, and $\delta := \underline{\delta} - \tau \gamma$, we have

$$\begin{aligned}
[A^T(Ax - b) + \beta x]_i &= \begin{cases} 0 & i \in \mathbb{K}_0 \\ 0 & i \in \mathbb{K}_1 \\ -\beta \theta_i t_i & i \in \mathbb{K}_2 \\ -\beta \theta_i t_i & i \in \mathbb{K}_3 \\ -\beta \theta_i t_i & i \in \mathbb{K}_4 \\ -\beta \theta_i s_i & i \in \mathbb{K}_5 \\ -\beta \theta_i s_i & i \in \mathbb{K}_6 \\ \beta \theta_i s_i - \beta \theta_i w_i & i \in \mathbb{K}_7 \end{cases} \\
&= \begin{cases} 0 & i \in \mathbb{K}_0 \\ 0 & i \in \mathbb{K}_1 \\ \beta x - \delta \beta \text{sign}(x_i) & i \in \mathbb{K}_2 \\ \beta x - \delta \beta \text{sign}(x_i) & i \in \mathbb{K}_3 \\ \beta x - \delta \beta \text{sign}(x_i) & i \in \mathbb{K}_4 \\ -(\delta \beta - w_i \beta) \text{sign}(x_i) & i \in \mathbb{K}_5 \\ -(\delta \beta - w_i \beta) \text{sign}(x_i) & i \in \mathbb{K}_6 \\ -(\delta \beta + 2w_i \beta) \text{sign}(x_i) & i \in \mathbb{K}_7 \end{cases}
\end{aligned}$$

Then apply polyhedra constraints on the previous expression. $\square$

Escaping point with $\gamma = 0$ with simplex method.

Our goal is to maximize $|\delta - \underline{\delta}| = |-\tau\gamma| = \gamma$ while keeping the polyhedral constraints valid and starting with green and red ones as basis.

$\max \gamma$

$$\begin{array}{llllll}
(A^T \underline{AM})_i u + \beta \underline{\theta}_i \underline{u}_i & & & & = (A^T b)_i & i \in \mathbb{K}_0 \\
(A^T \underline{AM})_i u + \beta \underline{\theta}_i \underline{u}_i & & & & = (A^T b)_i & i \in \mathbb{K}_1 \\
(A^T \underline{AM})_i u + \beta \underline{\theta}_i \underline{u}_i & + \beta \underline{\theta}_i t_i & & & = (A^T b)_i & i \in \mathbb{K}_2 \\
(A^T \underline{AM})_i u + \beta \underline{\theta}_i \underline{u}_i & + \beta \underline{\theta}_i t_i & & & = (A^T b)_i & i \in \mathbb{K}_3 \\
(A^T \underline{AM})_i u + \beta \underline{\theta}_i \underline{u}_i & + \beta \underline{\theta}_i t_i & & & = (A^T b)_i & i \in \mathbb{K}_4 \\
(A^T \underline{AM})_i u & + \beta \underline{\theta}_i s_i & & & = (A^T b)_i & i \in \mathbb{K}_5 \\
(A^T \underline{AM})_i u & + \beta \underline{\theta}_i s_i & & & = (A^T b)_i & i \in \mathbb{K}_6 \\
(A^T \underline{AM})_i u & - \beta \underline{\theta}_i s_i & + \beta \underline{\theta}_i w_i & & = (A^T b)_i & i \in \mathbb{K}_7 \\
u_i & + \tau\gamma & & - \underline{t}_i & = \underline{\delta} & i \in \mathbb{K}_0 \\
u_i & + \tau\gamma & & - \underline{t}_i & = \underline{\delta} & i \in \mathbb{K}_1 \\
u_i & + \tau\gamma & & + \underline{t}_i & = \underline{\delta} & i \in \mathbb{K}_2 \\
u_i & + \tau\gamma & & + \underline{t}_i & = \underline{\delta} & i \in \mathbb{K}_3 \\
u_i & + \tau\gamma & & + \underline{t}_i & = \underline{\delta} & i \in \mathbb{K}_4 \\
& \tau\gamma & + s_i & & + \underline{w}_i & = \underline{\delta} & i \in \mathbb{K}_5 \\
& \tau\gamma & + s_i & & + \underline{w}_i & = \underline{\delta} & i \in \mathbb{K}_6 \\
& \tau\gamma & + s_i & & + \underline{w}_i & = \underline{\delta} & i \in \mathbb{K}_7
\end{array}$$

$$u \geq 0 \quad \gamma \geq 0 \quad s \geq 0 \quad t \geq 0 \quad w \geq 0 \quad .$$

One can see that the constraint matrix is full rank, with $rank = 2n$.

Algorithm 1 Escaping $\gamma = 0$

- 1: Inputs $(\underline{\delta}, \underline{x})$.
 - 2: Perform Gauss-Jordan elimination with green and red as basis.
 - 3: Apply one Simplex method change of basis step.
 - 4: **if** there is no row with a nonnegative value on the γ column and a 0 value on the right hand side **then**
 - 5: We've found a point with $\gamma > 0$, and it is simplex optimal.
 - 6: $\bar{x} := \underline{M}u$, and $\bar{\delta} := \underline{\delta} - \tau\gamma$.
 - 7: **else**
 - 8: For each concerned row, do as follow,
$$\begin{aligned} \text{if } i \in \mathbb{K}_1 &\Rightarrow \mathbb{K}_1 := \mathbb{K}_1 \setminus \{i\}, \mathbb{K}_2 := \mathbb{K}_2 \cup \{i\} \\ \text{if } i \in \mathbb{K}_2 &\Rightarrow \mathbb{K}_2 := \mathbb{K}_2 \setminus \{i\}, \mathbb{K}_1 := \mathbb{K}_1 \cup \{i\} \\ \text{if } i \in \mathbb{K}_4 &\Rightarrow \mathbb{K}_4 := \mathbb{K}_4 \setminus \{i\}, \mathbb{K}_5 := \mathbb{K}_5 \cup \{i\} \\ \text{if } i \in \mathbb{K}_5 &\Rightarrow \mathbb{K}_5 := \mathbb{K}_5 \setminus \{i\}, \mathbb{K}_4 := \mathbb{K}_4 \cup \{i\} \end{aligned}$$
 - 9: and go to 1.
 - 10: **end if**
 - 11: Return $(\bar{\delta}, \bar{x})$.
-

Proposition 0.4.0.2 1. *At most two Gauss-Jordan eliminations and two Simplex Method change of basis steps are performed to obtain $\gamma > 0$.*
2. *$x^\beta(\delta)$ is a piecewise linear function with respect to $\delta \geq 0$.*

Proof:

1. ...
2. ...

□

Entire path direct solutions

This is a first approach that allow entire path solutions computation using Gauss-Jordan eliminations and Simplex Method. LARS like algorithm is also under progress!

Recall that

- $\underline{x} := 0$ is a solution for $(\mathbb{Q}^{\beta, \delta})$, for all $\delta \geq \delta^0 := \frac{1}{\beta} \|A^T b\|_\infty$.
- $\underline{x} := x^{Ridge^\beta}$ is a solution for $(\mathbb{Q}^{\beta, \delta})$,

$$\text{for all } \delta \leq \delta^{Ridge^\beta} := \min_{x_i^{Ridge^\beta} \neq 0} \left| x_i^{Ridge^\beta} \right|.$$

Algorithm 2 Full path computation $(\mathbb{Q}^{\beta,\delta})$, $\delta \geq 0$

```

1:  $\underline{\delta} := \delta^0$  and  $\underline{x} := 0$ ,  $\tau := +1$ ,
2:  $L := \{(\underline{\delta}, \underline{x})\}$ 
3: while  $\underline{\delta} > \min_{i=1,\dots,n} |\underline{x}_i|$  do
4:    $(\underline{\delta}, \underline{x}) :=$  output of Algorithm 1,
5:    $L := L \cup \{(\underline{\delta}, \underline{x})\}$ .
6: end while
7: return  $\hat{x}$ 

```

Algorithm 3 Full path computation $(\mathbb{Q}^{\beta,\delta})$, $\delta \geq 0$

```

1:  $\underline{\delta} := \delta^{Ridge^\beta}$  and  $\underline{x} := x^{Ridge^\beta}$ ,  $\tau := -1$ ,
2:  $L := \{(\underline{\delta}, \underline{x})\}$ 
3: while  $\underline{x} \neq 0$  do
4:    $(\underline{\delta}, \underline{x}) :=$  output of Algorithm 1,
5:    $L := L \cup \{(\underline{\delta}, \underline{x})\}$ .
6: end while
7: return  $\hat{x}$ 

```

We also implemented a parallel version using both Algorithm 2 and Algorithm 3, with a critical common section.

0.4.1 Cyclic coordinate descent for solving $(\mathbb{Q}^{\beta,\delta})$

Algorithm 4 A cyclic coordinate descent for $(\mathbb{Q}^{\beta,\delta})$

```

1: Precompute  $z_j := \sum_i^m A_{i,j}^2 = \|A_{:,j}\|_2^2$ 
2: Initialize  $\hat{x} := 0$ , (or something else)
3: while not converged do
4:   for  $j = 1, \dots, n$  do
5:     Compute  $\alpha_j := \sum_i^m A_{i,j}(b_i - \sum_{k \neq j} \hat{x}_k A_{i,k})$ ,
6:     and set

```

$$\hat{x}_j := \begin{cases} 0 & |\alpha_j| \leq \delta\beta, \\ \frac{\alpha_j - \delta\beta \text{sign}(\alpha_j)}{z_j} & \delta\beta < |\alpha_j| \leq \delta(\beta + z_j), \\ \frac{\alpha_j}{\beta + z_j} & \delta(\beta + z_j) \leq |\alpha_j|. \end{cases}$$

```

7:   end for
8: end while
9: return  $\hat{x}$ 

```

0.4.2 Interior point method

We will consider the convex quadratic form of $(\mathbb{Q}^{\beta,\delta})$, see Le Thi et al.

$$\begin{cases} \min & \|Ax - b\|_2^2 + \beta \|s + \delta v\|_2^2 - 2\beta\delta^2 e^T v \\ \text{s.t.} & -s_i \leq x_i \leq s_i, \ 0 \leq v_i, i = 1, \dots, n. \end{cases}$$

Logarithmic barrier for the bounds

$$\Phi(x, s, v) = - \sum_i^n \log(s_i + x_i) - \sum_i^n \log(s_i - x_i) - \sum_i^n \log(v_i).$$

$$\text{dom } \Phi = \left\{ \begin{pmatrix} x \\ s \\ v \end{pmatrix} : -s_i < x_i < s_i, \ 0 < v_i, \ i = 1, \dots, n. \right\}.$$

The central path consists of the unique minimizer $(x(t), s(t), v(t))$ of the convex function

$$\phi_t(x, s, v) = t \left[\|Ax - b\|_2^2 + \beta \|s + \delta v\|_2^2 - 2\beta\delta^2 e^T v \right] + \Phi(x, s, v).$$

Newton method is used to minimize ϕ_t , i.e, the search direction is computed as the exact solution to Newton system

$$H \begin{pmatrix} \Delta x \\ \Delta s \\ \Delta v \end{pmatrix} = -g.$$

The Hessian

$$H = \nabla^2 \phi_t(x, s, v),$$

and the gradient

$$g = \nabla \phi_t(x, s, v),$$

at the current iterate (x, s, v) .

...

0.4.3 Branch and bound method

Let us consider a branch LFZ with the partition

$$(A^L, A^F, A^Z) := A,$$

$$(x^L, x^F, x^Z) := x,$$

where x^Z are coordinates forced to zero, x^F coordinates forced to nonzero and x^L the ones concerned by the sparsity regularizer.

$$(\mathbb{P}_{LFZ}^{\beta,\delta}) \left\{ \begin{array}{ll} \min & \|A^L x^L + A^F x^F - b\|_2^2 \\ & + \beta \delta^2 \#F + \beta \|x^F\|_2^2 + \beta \delta^2 \|x^L\|_0 + \beta \|x^L\|_2^2 \\ \text{s.t.} & x \in \mathbb{R}^n. \end{array} \right.$$

is relaxed with

$$(\mathbb{Q}_{LFZ}^{\beta,\delta}) \left\{ \begin{array}{ll} \min & \|A^L x^L + A^F x^F - b\|_2^2 \\ & + \beta \delta^2 \#F + \beta \|x^F\|_2^2 + \beta \delta^2 \left(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2 \right)^{**} (x^L) \\ \text{s.t.} & x \in \mathbb{R}^n. \end{array} \right.$$

We first consider a simple branch and bound approach where dual values from $(\mathbb{Q}_{LFZ}^{\beta,\delta})$ will give us lower bounds. Corresponding primal solutions are also used as candidate solutions.

0.5 Sparse Principal Component

Consider a covariance matrix Q .

Writing $Q = A^T A$ and $b := A\alpha$, we can see that

$$\|Ax - b\|_2^2 = (x - \alpha)^T Q (x - \alpha),$$

$$A^T (Ax - b) = Q(x - \alpha)$$

and

$$A^T b = Q\alpha.$$

A first approach is to replace ELASTICNET engine by $(\mathbb{Q}^{\beta,\delta})$ in SPCA algorithm from Zou et al..

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